

UCI Census Income Classification — Day 2 Executive Report

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1. PREPROCESSING PLAN & TRADEOFF COMPARISONS

- Numerical Pipeline: SimpleImputer(strategy='median') → StandardScaler()
 - Why Median: Capital gain/loss have >90% zero inflation with extreme outliers (\$99,999). Median is robust against skew.
 - Why StandardScaler: Centers features to $\mu=0$, $\sigma=1$ for balanced gradient convergence and proportional L2 weight penalization.
- Categorical Pipeline: SimpleImputer(strategy='most_frequent') → OneHotEncoder(handle_unknown='ignore')
 - Why One-Hot: Nominal categories (e.g. occupation) have no natural order. 'handle_unknown=ignore' prevents inference crashes.
 - Alternatives Skipped: Ordinal Encoding (injects false geometric distance into linear models); Target Encoding (avoids leakage).

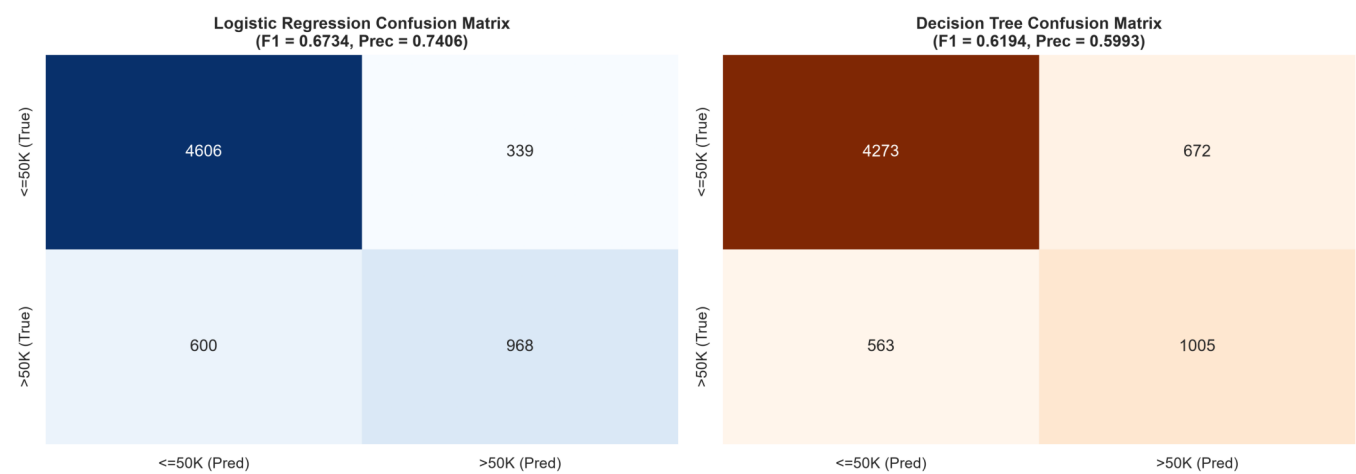
2. COMPREHENSIVE PERFORMANCE COMPARISON (6,513 HOLD-OUT TEST SET)

	Model / Baseline	Accuracy	Precision	Recall	F1-Score	ROC-AUC	PR-AUC
Day 1:	Majority Baseline (0)	0.7591	0.0000	0.0000	0.0000	0.2409	0.2409
Day 1:	Education Rule (3)	0.7367	0.4746	0.4879	0.4811	0.6582	0.3548
Day 2:	Decision Tree (Unp)	0.8104	0.5993	0.6409	0.6194	0.7525	0.4706
Day 2:	Logistic Regress	0.8558	0.7406	0.6173	0.6734	0.9078	0.7732

3. ERROR TYPE ANALYSIS & STRATEGIC BUSINESS IMPACT

- Logistic Regression Breakdown: True Negatives (4,606) | False Positives (339, 5.2%) | False Negatives (600, 9.2%) | True Positives (968).
- Decision Impact: False Negatives (600) are 1.77x more frequent than False Positives (339). In costly outbound wealth advisory, contacting low earners (False Positives) burns direct sales budget. High Precision (74.06%) ensures 3 out of 4 contacted leads convert.

Figure 1: Logistic Regression vs Decision Tree Confusion Matrices



UCI Census Income Classification — Day 2 Interpretability & Roadmap

Page 2: Model Interpretability, Generalization Diagnostics & Day 3 Selection

4. MODEL INTERPRETABILITY & OVERFITTING DIAGNOSTICS

- Logistic Regression Coefficients:
 - Top Positive: capital_gain (+2.25, 9.47x odds), Married-civ-spouse (+1.38, 3.97x odds), Wife (+1.07), Exec-managerial (+0.79).
 - Top Negative: Priv-house-serv (-1.38, 0.25x odds), Own-child (-1.17), Never-married (-1.04), sex_Female (-0.98).
- Decision Tree Overfitting Diagnosis:
 - Tree Depth: 52 levels | Leaves: 3,815. Train Score: 100% (F1=0.9999) vs Test Score: 81.04% (F1=0.6194) → Severe Overfitting!
 - Top Splits: Root partitions on Married-civ-spouse, followed by capital_gain ($>0.82\sigma$) and education_num ($>0.94\sigma$, Bachelors+).

5. MODEL SELECTION & DAY 3 ADVANCEMENT

- Primary Linear Model (Logistic Regression): Selected for further development due to outstanding generalization (Train F1: 0.6586 vs Test F1: 0.6734), zero overfitting, strong ROC-AUC (0.9078), and well-calibrated probabilistic confidence scores.
- Primary Non-Linear Candidate (Tree Ensembles): While single trees overfit, they discover powerful non-linear interactions. We will advance Tree methods into Random Forests and Gradient Boosted Trees (XGBoost/LightGBM) to control variance.
- Planned Day 3 Preprocessing Enhancements:
 - Log1p transforms on 'capital_gain' and 'capital_loss' + binary flag 'has_capital_gain'.
 - Seniority interaction terms (age * education_num and hours_per_week * age).
 - Macro-regional clustering for 41 native_country values to reduce one-hot dimensionality.

Figure 2: ROC Curves (AUC=0.9078) & Precision-Recall Curves (PR-AUC=0.7732)

